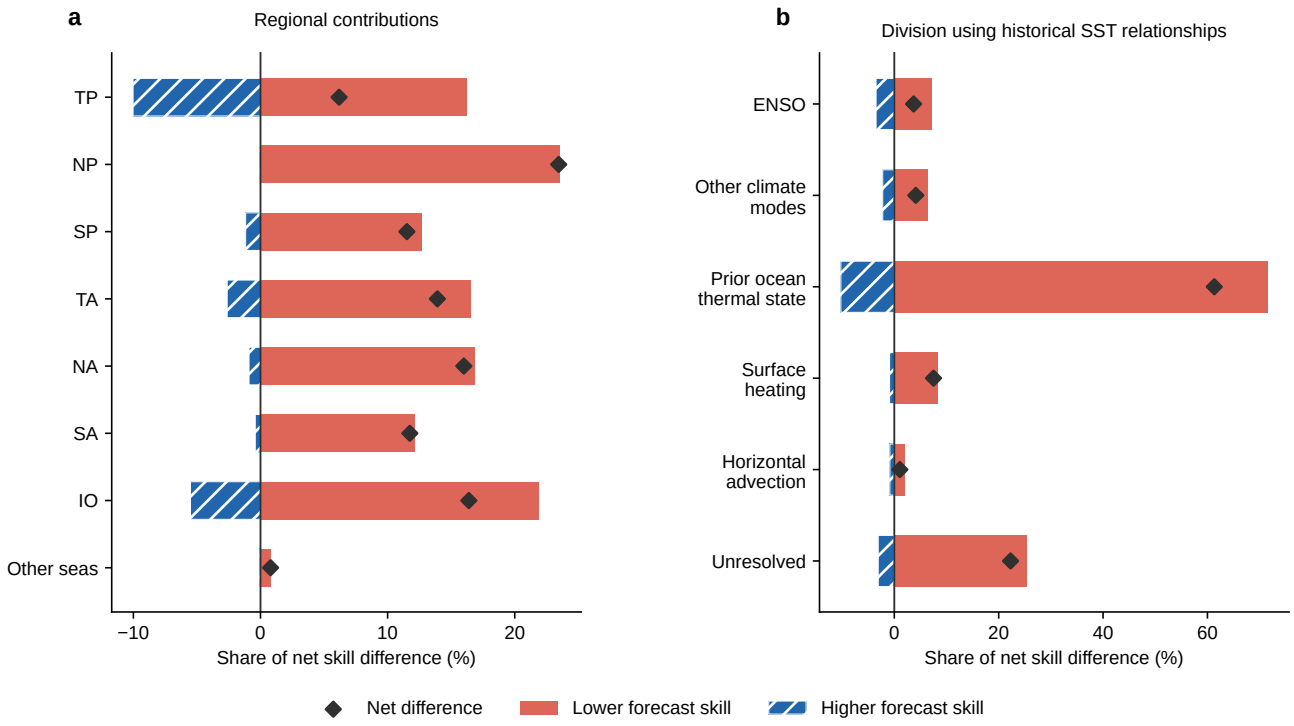
Supplementary Fig. 15 | Regional and component accounting of lower and higher forecast skill



a, Area-weighted regional contributions to the net difference between the globally expected AUC (0.7050185) and the local neighbourhood AUC in 2023–2024, matching main Fig. 4. Red bars sum cells below that reference; blue hatched bars subtract cells above it; black diamonds show the net contribution. b, The same signed amounts allocated using historical SST-association weights. The three local components are shown separately and their parent group is not added again. Both panels divide by the same net map difference, 0.1065144 AUC. Positive shares sum to 120.79% and negative shares to −20.79%, giving 100% net. Regions use ±23.5° boundaries; Other seas comprises evaluated cells outside the seven masks.

The map uses April 2023–September 2024, nine leads and 11° same-basin neighbourhoods. Its area-weighted local-AUC difference is not the monthly spatial-AUC deficit in main Fig. 1 (0.1090497). Component weights describe historical SST associations, not causal contributions of process forecast errors. Unresolved is the unallocated association weight. Supplementary Methods gives the signed-sum identity and distinguishes the two AUC metrics.